

Demonstrating the Equivalence of a Gravitational Field and an Accelerating Reference Frame

Finding Your Team Assignment & Coordinating with Your Team to Pick Up Equipment

- You should have groups assigned by Canvas.
- Contact your team members and designate someone to pick up your packet of equipment.
- Pick up equipment after class on Monday, Sept 16, or in Prof Breiviks's office, Wean 8319 during office hours from 2pm-3pm.
- You may do the project at your convenience, either during the week before the designated project day or on the designated days (Sept 18,20) during regular class time.

Making and Recording the Measurements

- All measurements should be done during our regularly scheduled class time unless agreed otherwise with your team.
- Decide as a team which TWO elevators you will use for measurements. You may use any elevators in academic buildings (but not in administrative buildings or the University Center).
- Provide enough information about the specific location of your elevators (e.g., directions, photos, room numbers, etc.).
- Check the “zero” on the force scale:
 - The larger hook is at the top end, fixed inside the scale, and the smaller hook is attached to the spring.
 - Hold the scale freely and adjust the zero if necessary by loosening the locking ring.
- Understand the scale readings:
 - The maximum force is 1.0 N.
 - Fine scale divisions are either 0.01 N or 0.02 N.
- Measure the gravitational force on a 50-gram mass at rest.
- Measure the apparent force on the object in two non-inertial reference frames (elevators):
 - Take the mass and scale on a ride in the elevator.
 - Measure the force at the instant the elevator starts to move up.

- Measure the force when the elevator comes to a stop while moving down.
- Only record readings that are greater than the reading at rest.

Comparing the Results

Answer the questions based on your measurements. Do you detect any significant difference between the Starting/UP and Stopping/DOWN measurements? Are the two elevators significantly different from each other?

Submitting the Results

Turn in one data sheet per team via file upload to Canvas on the scheduled due date. Return your equipment to Prof Breivik in class on Monday, Sept 23, or during office hours on Tuesday Sept 24.

Force Measurements in Accelerating Reference Frames

(Fill in this data sheet, and turn in the results on Canvas.)

Names of team members (print clearly):

Team Number: ____

Force on 50 gram object at rest = ____ N

Elevator 1 location

Building: _____

Specific Location: _____

Elevator 2 location

Building: _____

Specific Location: _____

Starting (Going Up)

1. ____ N
2. ____ N
3. ____ N
4. ____ N
5. ____ N

Stopping (Going Down)

1. ____ N
2. ____ N
3. ____ N
4. ____ N
5. ____ N

Starting (Going Up)

1. ____ N
2. ____ N
3. ____ N
4. ____ N
5. ____ N

Stopping (Going Down)

1. ____ N
2. ____ N
3. ____ N
4. ____ N
5. ____ N

Questions for thought...

- For a given elevator, do you detect any significant difference between the Starting/UP and Stopping/DOWN measurements?

- Are the two elevators that you tested significantly different from each other?